

Differential Equation

4. Linear differential Equation, Reducible Linear: Bernoulli's Form

The equation is given by $y' + p(x)y = Q(x)$

The solution is given by $y(x)e^h = \left(\int e^h Q dx + c \right), \quad h = \int p(x)dx,$

Where Integrating Factor is I.F. = e^h

Example: Solve the linear differential equation

$$y' - y = e^{2x}.$$

Solution: $p = -1, \quad Q = e^{2x}, \quad h = \int p dx = -x, \quad I.F. = e^{-x}$

$$y(x) = e^x \left[\int e^{-x} e^{2x} dx + c \right] = e^x [e^x + c] = ce^x + e^{2x}.$$

Example: Solve $y' + 2y = e^x (3 \sin 2x + 2 \cos 2x)$.

Solution: Here $p = 2, h = 2x$, so that

$$y = e^{-2x} \left[\int e^{2x} e^x (3 \sin 2x + 2 \cos 2x) dx + c \right] = e^{-2x} [e^{3x} \sin 2x + c] = ce^{-2x} + e^x \sin 2x.$$

Example: Initial value problem

Solve the initial value problem $y' + y \tan x = \sin 2x$, $y(0) = 1$.

Solution: Here $p = \tan x$, $Q = \sin 2x = 2 \sin x \cos x$

$$\int p dx = \int \tan x dx = \ln |\sec x|.$$

$$e^h = \sec x, \quad e^{-h} = \cos x, \quad e^h Q = (\sec x)(2 \sin x \cos x) = 2 \sin x,$$

$$\text{general solution of our equation is } y(x) = \cos x \left[2 \int \sin x dx + c \right] = c \cos x - 2 \cos^2 x.$$

From this and the initial condition, $1 = c \times 1 - 2 \times 1^2$; thus $c = 3$ and the solution of our initial value problem is $y = 3 \cos x - 2 \cos^2 x$.

Bernoulli equation. Verhulst equation. Logistic population model

$$y' + Py = Qy^n, \text{ Put } \frac{1}{y^{n-1}} = z$$

$$\frac{dz}{dx} + Pz = Q, \text{ Can be solved like linear Differential equation.}$$

Example: Solve the special Bernoulli equation, called the Verhulst equation

$$y' - Ay = -By^2 \quad (A, B > 0)$$

Solution: Here, $a = 2$, so that $u = y^{-1}$, and by differentiation and substitution of y' from

$$u' = -y^{-2} y' = -y^{-2} (-By^2 + Ay) = B - Ay^{-1},$$

$$u' + Au = B.$$

$$u = e^{-Ax} \left[\int B e^{Ax} dx + c \right] = e^{-Ax} \left[\frac{B}{A} e^{Ax} + c \right] = ce^{-Ax} + \frac{B}{A}.$$

$$\text{The general solution is, } y = \frac{1}{u} = \frac{1}{(B/A) + ce^{-Ax}},$$

